HDOC Day-65

Topic: Tower of Hanoi Using Recursion in C

Aim

To write a recursive C program to solve the Tower of Hanoi problem for n disks and display every move.

Algorithm

1. Read the number of disks n.
2. If n = 1, move the disk from source to destination.
3. Otherwise, move n - 1 disks from source to auxiliary peg.
4. Move the largest disk from source to destination peg.
5. Move n - 1 disks from auxiliary to destination peg.

C Program

```
#include <stdio.h>

void hanoi(int n, char source, char dest, char aux) {
    if (n == 1) {
        printf("Move disk 1 from %c to %c\n", source, dest);
        return;
    }
    hanoi(n - 1, source, aux, dest);
    printf("Move disk %d from %c to %c\n", n, source, dest);
    hanoi(n - 1, aux, dest, source);
}

int main() {
    int n;
    printf("Enter number of disks: ");
    scanf("%d", &n);
    if (n <= 0) {
        printf("Please enter a positive number.\n");
        return 0;
    }
    hanoi(n, 'A', 'C', 'B');
    return 0;
}
```

Sample Output (n = 3)

```
Enter number of disks: 3
Move disk 1 from A to C
Move disk 2 from A to B
Move disk 1 from C to B
Move disk 3 from A to C
Move disk 1 from B to A
Move disk 2 from B to C
```

```
Move disk 1 from A to C
```

Explanation

The recursive function moves smaller disks out of the way, transfers the largest disk, and then moves the smaller disks onto it. The minimum number of moves is $2^n - 1$; for 3 disks, 7 moves are required.

Conclusion

The Tower of Hanoi problem was successfully solved using recursion, and all disk movements were displayed.